DATABASE MANAGEMENT SYSTEMS

PURPOSE OF DATABASE SYSTEM

File-processing system has a number of major disadvantages:

- **Data redundancy and inconsistency**

  Same information may be duplicated in several places.

  All copies may not be updated properly.

- **Difficulty in accessing data**

  Probability of writing a new application program to satisfy an unusual request.

PURPOSE OF DATABASE SYSTEM

- **Data Isolation**
  - ➢ Data are scattered in various files.
  - ➢ Data in different formats.
  - ➢ Difficult to write new application programs.
- **Integrity problems**
  - ➢ The data values stored in the database must satisfy certain types of consistency constraints.
  - ➢ Difficult to enforce or to change constraints with the file-processing approach.

PURPOSE OF DATABASE SYSTEM

- **Atomicity problems**
  - ➢ Computer system is subject to failure.
  - ➢ The data be restored to the consistent state that existed prior is subject to failure.
- **Concurrent-access anomalies**
  - ➢ Want concurrency for faster response time.
  - ➢ Need protection for concurrent updates.

PURPOSE OF DATABASE SYSTEM

- **Security problems**

Every user of the system should be able to access only the data they are permitted to see.

These problems led to the development of database management systems.

DATA

- They are the raw facts that can be recorded and that have implicit meaning.

DATABASE

- It is the collection of related data with some inherent meaning.
- It is designed, built and populated with data for specific purpose.

FILE SYSTEM

- It is the method for storing and organizing computer files and the data they contain to make it easy to find and access them.
- It is controlled by the operating system.

DATABASE MANAGEMENT SYSTEM

- It is a collection of programs that enables to store, retrieve, define and manage data in a database.
- It is a general purpose software system that facilitates the process of defining, constructing and manipulating databases for various applications.

DATABASE MANAGEMENT SYSTEM

- **Defining a database** involves specifying the data types, structures, and constraints of the data to be stored in the database.
- **Constructing the database** is the process of storing the data on some storage medium that is controlled by the DBMS.
- **Manipulating a database** includes functions such as querying the database to retrieve specific data, updating the database to reflect changes and generating reports from the data.
- **Sharing a database** allows multiple users and programs to access the database simultaneously.

BASIC TERMINOLOGIES

Database system:

- Database and DBMS together known as the database system.

Database schema:

- The description/logical design of the database is called the database schema.

Schema diagram:

- The displayed schema is called a schema diagram.

Instance:

- It is the snapshot of the data in the database at a given instant of time.

DATA INDEPENDENCE

- Data Independence is defined as a property of DBMS that helps us to change the database schema at one level of a database system without requiring to change the schema at the next higher level.
- Data independence helps us to keep data separated from all programs that make use of it.
- It helps to improve the quality of data.